

effects will be realized earlier than is currently estimated under the existing MDI timeline because this portion of the MDI Rules is currently not set to be implemented until the end of the implementation timeline for the MDI Rules. Further, because the first steps of the timeline for the MDI Rules have not been accomplished,¹⁵⁶⁷ and the Commission is uncertain when exactly the round lot definition otherwise will be implemented, the degree of the effect of the acceleration is unknown.¹⁵⁶⁸

The Commission recognizes that the earlier implementation of the round lot definition could affect the tiered tick structure by sooner increasing the number of stocks subject to a minimum pricing increment of less than \$0.01, but the Commission does not expect this effect to be substantial. Specifically, a mechanically tighter NBBO will reduce the time-weighted average quoted spread used to determine the appropriate tick increment for stocks priced greater than \$250. However, higher-priced stocks also tend to have higher spreads that are unlikely to narrow enough for the amendments to result in a smaller minimum pricing increment.¹⁵⁶⁹

As discussed in the Proposing Release¹⁵⁷⁰, the Commission also recognizes that both the reduction in tick size and accelerating the definition of round lot will reduce the depth of liquidity at the NBBO. These effects might amplify each other in a small set of stocks. A reduction in tick size will spread liquidity across more price levels, while the implementation of

¹⁵⁶⁷ See supra notes 74-78 and accompanying text section for a discussion of the delays.

¹⁵⁶⁸ See supra section V.B.1 for a discussion of the factors that affect when MDI will be implemented and a discussion of an estimate of the proposed acceleration of at least two years after the Commission's approval of the plan amendment(s) required by rule 614(e).

¹⁵⁶⁹ In the MDI Rules the Commission estimated an average reduction in quoted spreads, conditional on the round lot definition resulting in a reduction of roughly 15% for stocks priced \$250-\$1,000 and 28% for stocks priced \$1,000-\$10,000. Given the average quoted spread of \$0.35 for stocks priced \$250-1,000 and \$2.90 for stocks priced \$1,000-\$10,000 the expected mechanical reductions are likely not sufficient to reduce the spreads of many of these stocks to the point where they would qualify for a lower tick size in this proposal. See MDI Adopting Release, supra note 10, at 18743.

¹⁵⁷⁰ See generally, Proposing Release, supra note 11.

the round lot definition will result in displaying smaller quotes at the NBBO. The amendments could result in this effect being amplified for stocks that trade above \$250 with spreads narrower than \$0.015 as these stocks will receive both smaller tick and smaller round lot sizes. The number of such affected stocks is likely very small.¹⁵⁷¹ The reduction in depth at the NBBO will temporarily reduce the information about liquidity available in the market for market participants who rely on public data feeds. However, the eventual inclusion of the depth of book information in consolidated market data under the MDI Rules, once implemented, will render this effect temporary. At that point in time, consolidated market data is expected to contain depth information at more price points, which will largely counteract the effects of a reduction in displayed depth from the implementation of the round lot definition and even from a reduction in tick size.

Multiple commenters further discussed the potential interaction of the reduction in tick size and the MDI round lot definition. One commenter stated that the resulting reduction in depth at the NBBO would make the NBBO less relevant and subject to more instability: “With a lower notional value earning protected status at the NBBO, even accessing 100 shares of liquidity would likely move a stock in one of the new tiers’ multiple price levels. Furthermore, these smaller notional amounts reduce the risk taken to queue jump displayed orders by placing slightly more aggressively priced orders ahead of them.”¹⁵⁷² A separate commenter stated that the reduced liquidity at the NBBO will require investors executing large orders to, “sweep across multiple market centers, exposing them to greater execution risk. Together, these changes would

¹⁵⁷¹ See section V.B.3.b for analysis identifying such stocks. In particular, see supra note 801 identifying only two stocks—both highly liquid—that would have qualified for both the tick reduction and a reduction in the round lot as of Nov. 30, 2023.

¹⁵⁷² See RBC Letter at 5. Queue jumping in this context is synonymous with pennyng. See supra note 994.

reduce the depth at the NBBO, leaving it subject to greater volatility and, in turn, reducing reliability and execution quality for retail investors.”¹⁵⁷³

The Commission acknowledges that these are possibilities, but the interaction of the reduction in tick size and the MDI round lot definition is not expected to have a material impact on the NBBO of affected stocks. In order for a stock to be impacted by both the new round lot categories and the smaller tick size, it would need to have a price over \$250 with a spread below \$0.015; this implies that the percentage spread must be below 0.006%.¹⁵⁷⁴ To put this in perspective, consider the sample of all stock-days in 2023.¹⁵⁷⁵ For each stock-day, divide its TWAQS by its price to measure its percentage spread. Only 1% of this sample has a percentage spread below 0.015%—i.e., the first percentile of the sample’s percentage spread is 2.5 times higher than the percentage spread of the stocks affected by both the new round lot categories and the smaller tick size.¹⁵⁷⁶ This implies that the affected stocks are exceptionally liquid—they are well within the first (i.e., most liquid) percentile when liquidity is measured using percentage spread. The exceptional liquidity of the affected stocks will likely protect their NBBO from material deterioration.

The Commission does acknowledge that the amendments will increase the likelihood that a 100-share order will walk the book for stocks affected by both a reduction in the tick and the

¹⁵⁷³ See Virtu Letter II at 8.

¹⁵⁷⁴ The percentage spread measures the cost of liquidity, measured here as the spread, as a fraction of the execution price. A lower percentage spread indicates that transaction costs for liquidity demanders are a smaller fraction of the execution price. Here, $\$0.015/\$250 = 0.00006 = 0.006\%$.

¹⁵⁷⁵ A symbol-day is the unique pair of a stock symbol and a date. For example, one observation is AMZN on December 7, 2023; a second observation is AMZN on December 8, 2023; a third is AAPL on December 7, 2023, etc.

¹⁵⁷⁶ This statistic is computed using all symbol-days in WRDS intra-day indicators for the year 2023. The sample has 2.3 million observations.

round lot definition;¹⁵⁷⁷ however, the high price (over \$250 per share) of the affected stocks implies that the notional amount of such an order will be substantially larger than the notional amount of a 100-share order for a typical stock unaffected by the MDI Rules round lots.

Therefore, a round lot under the new definition will continue to reflect a meaningful notional amount even with a reduced number of shares,¹⁵⁷⁸ thereby ensuring that regulatory protections for round lots—such as those governing the display, dissemination, and protection of orders under Rules 602, 604, and 611—continue to focus on orders of significant size. Likewise, the amendments will increase the likelihood that particularly large orders may need to sweep across multiple market centers; however, fixing the notional amount of an order, a high-priced stock will require fewer shares, which reduces the need to sweep across multiple market centers.

Therefore, the amendments will continue to protect a meaningful notional amount at the NBBO after the round lot size is reduced for these high-priced stocks. Similarly, the amendments will make it incrementally easier for traders to queue jump—*i.e.*, penny—protected orders in affected stocks, but pennyng will remain difficult due to the relatively high price of the affected stocks. That is, a trader would still need to commit a relatively high notional amount to jump the queue with a protected order. Finally, the concern with pennyng is that a market participant can jump the queue by posting economically trivial price improvement; for the stocks affected by both the

¹⁵⁷⁷ A marketable order “walks the book” if the size of the order is larger than the amount of liquidity available at the best price at a market center; the order must therefore execute against liquidity at multiple price points within the limit order book.

¹⁵⁷⁸ For example, the notional amount reflected by a round lot of 40 shares will generally be greater than \$10,000 (this is because a stock’s price must generally be greater than \$250 to be assigned a round lot of 40 shares, and $40 \times \$250 = \$10,000$). If a stock has a round lot of 100 shares, a round lot will only reflect \$10,000 of notional if the stock price is at least \$100 (so that $100 \times \$100 = \$10,000$); many stocks with a round lot of 100 shares do not have a price greater than \$100. Therefore, the lower round lot for high-priced stocks will continue to reflect a notional amount that is at least as high as the notional reflected in round lots for stocks with more common prices.

tick reduction and the round lot definition, however, posting a protected order that improves on the NBBO is likely to provide meaningful price improvement. To jump the queue with a protected order, a trader would need to post an order that is: (1) priced better than existing orders by \$0.005 (which is large relative the stock’s typical spread of under \$0.015), and (2) with a \$10,000 notional value¹⁵⁷⁹ (which is larger than the notional value required to queue jump for stocks priced under \$100). Such an order would thereby require the trader to offer price improvement that is economically large relative to the costs of trading the stock, and relative to the notional amount required to jump the queue in most other stocks. Therefore, the Commission continues to expect that the acceleration of the round lot definition will protect a meaningful amount of liquidity at the NBBO for stocks receiving the tick reduction.¹⁵⁸⁰

One commenter encouraged the Commission to “comprehensively review its proposed changes to tick sizes, access fees and round lots to better evaluate how these changes together would impact liquidity.”¹⁵⁸¹ As discussed in the preceding paragraphs, the Commission expects that a very small number of stocks (two as of Nov. 30, 2023¹⁵⁸²) would be subject to both a change in round lot size and tick size because very few stocks have both a price above \$250 (to qualify for a reduced round lot) and a TWAQS below \$0.015 (to qualify for the tick reduction).

¹⁵⁷⁹ The new round lot definitions are structured so that they protect \$10,000 of notional value. See Table 1 of MDI Adopting Release, supra note 10.

¹⁵⁸⁰ For example, suppose a trader wants to incrementally improve the NBBO by jumping ahead of a resting protected order. If the stock has a tick of \$0.005 and a round lot of 40 shares, then the trader must improve the price by \$0.005 and post an order with a notional value of at least \$10,000, see id. The cost of queue-jumping is therefore at least \$50 ($0.005 \times \$10,000$). Now consider a stock with a tick of \$0.01, a round lot of 100 shares, and a price of \$30 per share. The cost of queue-jumping for this stock is only \$30 ($0.01 \times 100 \times \30). Therefore, stocks that receive both a tick reduction and a reduced round lot under these amendments are not expected to experience more queue jumping—and consequent deterioration of the NBBO—compared to stocks that retain the penny tick and the 100-share round lot.

¹⁵⁸¹ See UBS Letter at 10.

¹⁵⁸² See supra note 801 identifying only two stocks—both highly liquid—that would have qualified for both the tick reduction and a reduction in the round lot as of Nov. 30, 2023.

The exceptional liquidity of the stocks with both these characteristics is unlikely to be materially affected by the interaction of the tick reduction and the reduction in the round lot. With respect to the reduction in the access fee cap, the Commission expects effects of that change to be independent of the effects of the round lot definition for two reasons. First, the reduction in the access fee cap is unlikely to affect a stock's round lot size. This is because the reduction in the access fee cap—to the extent that it affects quoted prices as discussed in sections VII.B.3 and VII.2—is not expected to move quoted prices by more than one tick. Round lots, on the other hand, are determined by whether a stock is priced above \$250, \$1,000, or \$10,000; the probability that the reduction in the access fee cap affects a stock's round lot assignment is therefore miniscule. Second, the reduction in the access fee cap and the reduction in the round lot are expected to have separate but unrelated effects on the NBBO. Stocks that receive a round lot less than 100 shares are expected to have a narrower NBBO because the new round lot definition will include quotes at better prices in core data that were previously excluded from being reported because they consisted of too few shares.¹⁵⁸³ Access fees do not affect the existence of these better priced quotes with fewer shares. The reduction in the access fee cap is expected to put upward pressure on quoted spreads and therefore widen the NBBO;¹⁵⁸⁴ this effect operates at a per-share level because fees and rebates are assessed per-share, making the number of shares in a round lot irrelevant. Therefore, the Commission does not expect the round lot definition to interact with the reduction in the access fee cap.

For institutions that do not purchase proprietary feeds, the MDI Rules once implemented will result in the display of five levels of depth-of-book in NMS market data. To the extent that

¹⁵⁸³ See MDI Adopting Release, *supra* note 10, at 18742.

¹⁵⁸⁴ See section VII.B.3 for a discussion of the effect that fees and rebates have on quoted spreads.

the amendments result in liquidity spread out across more price levels due to the round lot reduction,¹⁵⁸⁵ then these changes would reduce the value of these NMS market data. However, the high price of stocks affected by the round lot reduction implies that the amount of visible notional liquidity will remain high relative the notional liquidity visible for a typical stock unaffected by the MDI round lots.¹⁵⁸⁶

One commenter stated that, “the Commission failed to note how much actual volume takes place in any of the three proposed [round lot] tiers and what challenge, if any, changes to round lot definitions would address.”¹⁵⁸⁷ In the MDI Rules, the Commission estimated that approximately 1% of stocks, 3% of share volume, and 30% of dollar volume will be affected by the new round lot tiers.¹⁵⁸⁸ The Commission also discussed in the MDI Rules the effect of changing the round lot size on transparency and execution quality.¹⁵⁸⁹

The commenter further suggested that the implementation of the round lot definition could cause confusion among retail investors: “Currently, retail customers, especially those trading options, understand one option contract represents one hundred shares. Frequently, it is simply referred to as a ‘round lot.’ Changes in round lot sizes will most certainly create confusion in this area for retail investors.”¹⁵⁹⁰

¹⁵⁸⁵ See supra note 1348 and surrounding discussion on the effect that the tick reduction is expected to have on the value of MDI NMS market data.

¹⁵⁸⁶ See supra note 1578 for an example indicating that the amount of notional liquidity reflected in a round lot will generally be higher for stocks receiving a reduction in the round lot size than stocks that retain a round lot size of 100 shares (due to the higher price of stocks receiving a round lot reduction).

¹⁵⁸⁷ See Tastytrade Letter at 22.

¹⁵⁸⁸ See MDI Adopting Release, supra note 10, at 18743 Table 4.

¹⁵⁸⁹ See supra note 1563.

¹⁵⁹⁰ See Tastytrade Letter at 22.

The Commission acknowledges that there may be a learning curve associated with the new round lot definition. However, as discussed in the MDI Adopting Release, investor confusion will be temporary for four reasons. First, market participants already regularly trade in increments other than 100 shares.¹⁵⁹¹ Second, most NMS stocks will continue to have a round lot of 100 shares. Third, core data will be distributed with the size of the NBBO and best quotes in shares rather than in the number of round lots. Fourth, broker-dealers and other market participants will modify or develop their systems to automatically keep track of the round-lot changes.¹⁵⁹² Further, any confusion from the accelerated round lot definition would have occurred eventually under the original MDI timeline, so the incremental effect of MDI acceleration on investor confusion is minimal.

Other commenters expressed concerns about monthly updates to stocks' round lots. Each update requires market participants to "reconfigure their investment platforms and trading systems to make any modifications effective."¹⁵⁹³ The same commenter pointed out that, under the proposal, round lots would be assigned at discrepant intervals from tick assignments. Commenters also stated that these system updates may increase complexity and operational risk, and further contribute to investor confusion.¹⁵⁹⁴

While any periodic system update can pose a risk of glitches, the amendments assign round lots and tick sizes on the same schedule—every six months in May and November.

¹⁵⁹¹ See supra note 791 and surrounding text for a discussion on the interaction of the new round lot definition and options trading. It is unlikely that the new round lot definition will confuse retail investors trading in options, partially because options markets already have standard contracts on stocks with a round lot size less than 100 shares.

¹⁵⁹² See MDI Adopting Release, supra note 10, at 18745.

¹⁵⁹³ See BlackRock Letter at 9-10.

¹⁵⁹⁴ See SIFMA Letter II at 34.

Syncing the updates like this will reduce costs relative to the monthly round lot updates in the baseline by reducing the number of times that firms are required to “open the hood” of trading systems. To further reduce these costs and provide opportunity for industry testing, the adopted amendments incorporate a one-month gap between evaluation periods and the implementation of updated round lots and tick sizes. It is possible that the amendments to the round lot definition—i.e., the less frequent evaluation periods and the lag between evaluation and implementation—may cause a stock’s round lot to be less reflective of its price than would have otherwise been the case under the original MDI Rules round lot definition (e.g., if a stock’s price falls after the evaluation period, it may be assigned to a round lot that is too low for the next six months). This imprecision in round lot assignment, however, is unlikely to significantly reduce the benefits of the MDI Rules for two reasons. First, to the extent that a stock is assigned a round lot based on stale information, this assignment will be corrected within six months at the next evaluation date. Second, given the significant distance between the round lot thresholds (i.e., \$250, \$1,000, and \$10,000), any deviation in a stock’s round lot as a result of these amendments is likely to be due to stocks that are near a threshold; for these stocks, the cost of being including in the next smallest or largest tier is likely to be small.

Finally, one commenter suggested alternative price thresholds for the round lot definition; these thresholds would result in five round lot tiers.¹⁵⁹⁵ The Commission continues to believe, as stated in the MDI Adopting Release, that a five-tiered approach is unnecessarily complex, and that the adopted tiers promote a smoother transition to a price-based round lot structure.¹⁵⁹⁶

¹⁵⁹⁵ See Pragma Letter at 9.

¹⁵⁹⁶ See MDI Adopting Release, supra note 10, at 18618. The price-based round lot structure ensures that there is \$10,000 of notional value protected under the new round lot definitions. See supra note 1579.

b. Including Odd-Lots in NMS Data

As discussed in the Proposing Release,¹⁵⁹⁷ the acceleration of the implementation of the MDI Rules that expand the NMS data to include odd-lot information inside the NBBO will result in sooner realizing some, but not all, economic effects of this aspect of the MDI Rules.¹⁵⁹⁸ The odd-lot information could be useful to consumers of SIP data that could use it to make better inferences about market conditions, thereby leading to investment decisions that more fully reflect market conditions and increased market efficiency. This odd-lot information could also lessen the effect of a reduction in displayed depth at the NBBO resulting from either a smaller tick size or a smaller round lot. Specifically, expediting inclusion of odd-lot data will allow individual investors whose broker-dealers subscribe to the data to visually monitor the market sooner than they would otherwise.¹⁵⁹⁹

Multiple commenters remarked on the growing importance of odd-lot activity for the overall equities market. One commenter stated that odd-lots “provide a meaningful source of liquidity across all trading sessions and stocks, representing 54.8% of all trades in the U.S. financial markets, up from 43% at the beginning of 2020.”¹⁶⁰⁰ Similarly, another commenter stated that including odd-lot information in consolidated market data will “improve transparency and increase the usefulness of the consolidated tape given the growing prevalence of market

¹⁵⁹⁷ See Proposing Release, supra note 11, section V.D.5.

¹⁵⁹⁸ See MDI Adopting Release, supra note 10, section V.C.1(c)(i), for the full discussion of the effects of including odd-lot information inside the NBBO in its definition of core data. Also, the MDI Rules do not require that the competing consolidators to disseminate odd-lot information, but the Commission anticipated in the MDI Adopting Release that at least one would do so. The requirement that the exclusive SIPs disseminate odd-lot information helps ensure that the economic effects of the acceleration of the MDI Rules occur. See infra section VII.D.5.c for a discussion of the costs to the exclusive SIPs.

¹⁵⁹⁹ See MDI Adopting Release, supra note 10, section V.C.1(c)(i).

¹⁶⁰⁰ See Cboe Letter II at 10.